

Machine Learning-Based Prediction of Mortality Risk in Patients with Heart Failure: A Comparative Study of Logistic Regression, Random Forest, and XGBoost

Abstract

Background: Heart failure is a major cardiovascular disease associated with high mortality. Accurate prediction of mortality risk during follow-up is essential for early clinical intervention and personalized management.

Objective: This study aimed to identify important clinical factors associated with mortality and develop machine learning models for predicting the probability of death among patients with heart failure.

Methods: A clinical dataset containing 299 patients with heart failure and 13 clinical variables was analyzed. Statistical analyses, including independent samples t-test and Spearman correlation analysis, were performed to identify mortality-associated factors. Five clinically relevant variables were selected as model inputs. Three machine learning algorithms, including Logistic Regression, Random Forest, and XGBoost, were developed and evaluated using an 80% training set and a 20% testing set. Model performance was assessed using Accuracy, Precision, Recall, F1-score, and the area under the receiver operating characteristic curve (AUC).

Results: The identified important factors associated with mortality included follow-up time, serum creatinine, ejection fraction, age, and serum sodium. Among the evaluated models, Random Forest achieved the highest AUC (0.900), while XGBoost achieved the highest accuracy (0.850) and F1-score (0.727).

Conclusion: Machine learning approaches can effectively predict mortality risk in patients with heart failure. The proposed framework provides a potential tool for individualized mortality probability estimation and clinical risk assessment.

1 Introduction

Heart failure (HF) is a complex clinical syndrome characterized by impaired cardiac function and remains one of the leading causes of morbidity and mortality worldwide. Despite advances in medical treatment, patients with heart failure continue to experience substantial risks of hospitalization and death. Therefore, early identification of high-risk patients is important for improving clinical outcomes.

Traditional statistical approaches, such as logistic regression, have been widely used in cardiovascular risk prediction due to their interpretability. However, the complex interactions among clinical variables may limit the predictive performance of conventional methods. Recently, machine learning algorithms have shown potential in clinical prediction tasks by capturing nonlinear relationships between multiple patient characteristics.

In this study, we developed a machine learning-based mortality prediction framework for patients with heart failure. First, statistical analysis was performed to identify mortality-associated clinical factors. Then, selected features were used to construct and compare three machine learning models: Logistic Regression, Random Forest, and XGBoost. Finally, the optimal prediction framework was applied to estimate the probability of mortality during the follow-up period for individual patients.

2 Methods

2.1 Dataset

This study utilized the Heart Failure Clinical Records Dataset, which contains clinical information from 299 patients with heart failure. The dataset includes 13 clinical variables, including demographic characteristics, laboratory measurements, and clinical history indicators. The prediction target was *DEATH_EVENT*, where 1 indicates death during follow-up and 0 indicates survival.

2.2 Risk Factor Identification

To identify clinical factors associated with mortality, independent samples t-test was first applied to compare differences between survival and death groups. Furthermore, Spearman correlation analysis was conducted to evaluate the relationship between clinical variables and mortality outcome.

The correlation coefficient between variables and mortality was calculated as:

$$\rho = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

where ρ represents the Spearman correlation coefficient, X represents clinical variables, and Y represents the mortality outcome.

2.3 Feature Selection

Based on statistical significance from the t-test and correlation strength from Spearman analysis, five clinically relevant variables were selected as model inputs:

$$X = (\text{time}, \text{serum_creatinine}, \text{ejection_fraction}, \text{age}, \text{serum_sodium})$$

These variables represented important aspects of disease progression, renal function, cardiac function, demographic characteristics, and electrolyte balance.

2.4 Machine Learning Models

Three machine learning algorithms were developed:

- Logistic Regression
- Random Forest
- XGBoost

Logistic Regression was used as a baseline probabilistic model. The predicted mortality probability was calculated as:

$$P(Y = 1|X) = \frac{1}{1 + \exp[-(\beta_0 + \sum_{i=1}^n \beta_i X_i)]}$$

where $Y = 1$ represents mortality during follow-up and X_i represents selected clinical features.

2.5 Model Evaluation

The dataset was randomly divided into training and testing sets with a ratio of 80% to 20%. Model performance was evaluated using Accuracy, Precision, Recall, F1-score, and receiver operating characteristic (ROC) analysis with AUC calculation.

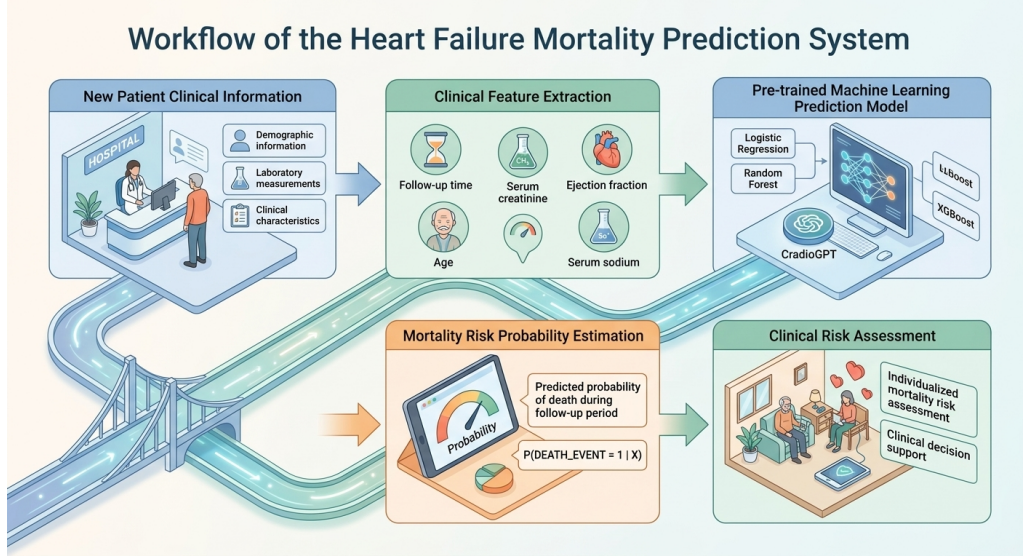


Figure 1: Workflow of the proposed heart failure mortality prediction system.

3 Results

3.1 Baseline Characteristics of Patients

A total of 299 patients with heart failure were included in this study. The baseline characteristics of the study population are summarized in Table ??.

The mean age of the patients was 60.83 ± 11.89 years. The average ejection fraction was $38.08 \pm 11.83\%$, and the mean serum creatinine level was 1.39 ± 1.03 mg/dL. Among all patients, 96 (32.11%) experienced death during the follow-up period.

Table 1: Baseline characteristics of patients with heart failure.

Variable	Mean $\pm$ SD / n(%)
Age (years)	60.83 ± 11.89
Anaemia	129 (43.14%)
Creatinine phosphokinase	581.84 ± 970.29
Diabetes	125 (41.81%)
Ejection fraction (%)	38.08 ± 11.83
High blood pressure	105 (35.12%)
Platelets	263358.03 ± 97804.24
Serum creatinine	1.39 ± 1.03
Serum sodium	136.63 ± 4.41
Male sex	194 (64.88%)
Smoking	96 (32.11%)
Follow-up time (days)	130.26 ± 77.61
Death event	96 (32.11%)

3.2 Identification of Mortality-Associated Factors

Independent samples t-test and Spearman correlation analysis were performed to identify clinical variables associated with mortality.

The correlation heatmap of the top 10 factors associated with *DEATH_EVENT* is shown in Figure ??.

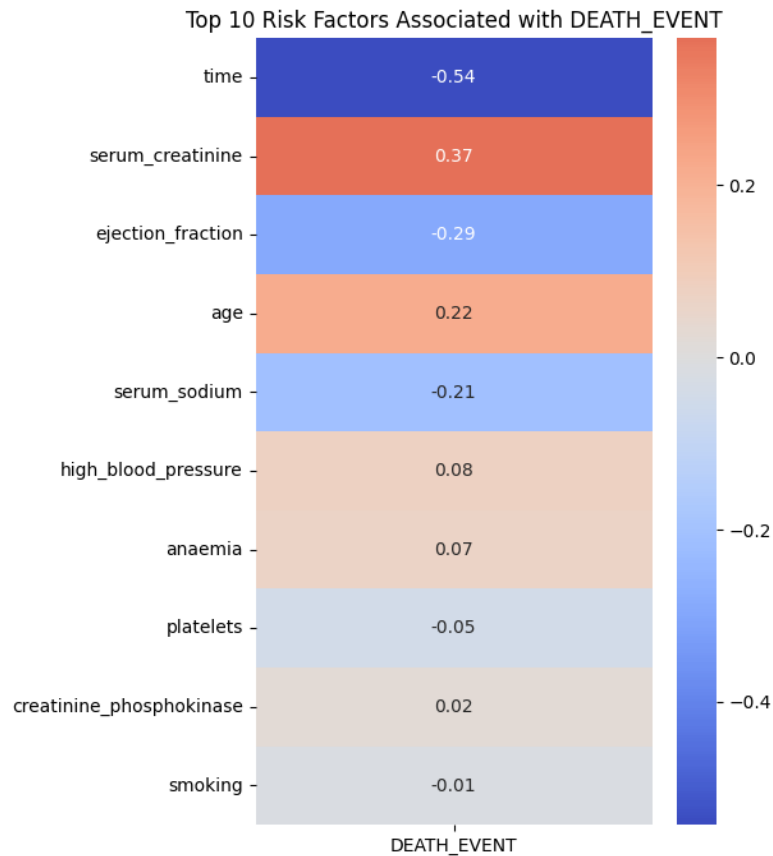


Figure 2: Spearman correlation heatmap of clinical variables associated with *DEATH_EVENT*.

Among all evaluated variables, follow-up time showed the strongest correlation with mortality outcome. Serum creatinine demonstrated a positive association with mortality, indicating that impaired renal function may be related to poor clinical outcomes. In contrast, ejection fraction and serum sodium showed negative correlations, suggesting that reduced cardiac function and electrolyte imbalance may contribute to increased mortality risk.

Based on the statistical analysis, five variables were selected as predictors:

- time
- serum creatinine
- ejection fraction
- age
- serum sodium

3.3 Comparison of Machine Learning Models

Three machine learning models were developed using the selected clinical features. The performance comparison results are summarized in Table ??.

Table 2: Performance comparison of different machine learning models.

Model	Accuracy	Precision	Recall	F1-score	AUC
Logistic Regression	0.833	0.800	0.632	0.706	0.890
Random Forest	0.817	0.750	0.632	0.686	0.900
XGBoost	0.850	0.857	0.632	0.727	0.873

Among the three models, XGBoost achieved the highest accuracy and F1-score, indicating better classification performance in identifying patients with mortality events. Random Forest obtained the highest AUC value (0.900), suggesting superior discrimination ability between survival and mortality groups. Logistic Regression showed competitive performance while maintaining high interpretability.

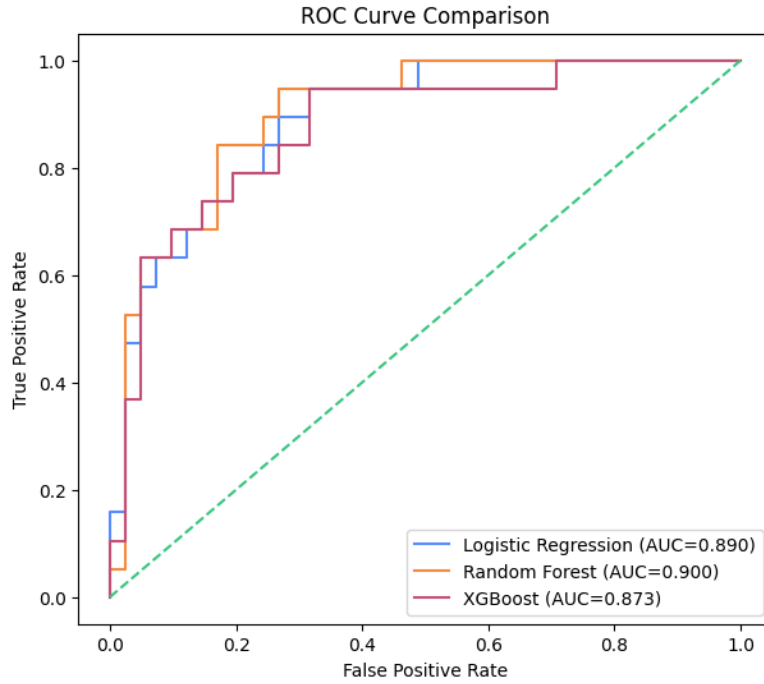


Figure 3: ROC curves of Logistic Regression, Random Forest, and XGBoost models for mortality prediction.

3.4 Clinical Application Workflow

After comparing different machine learning models, the developed prediction system can be applied to individual patients. The workflow consists of inputting clinical characteristics, processing patient information through the selected prediction model, and generating an estimated probability of mortality during follow-up.

The workflow of the proposed prediction system is illustrated in Figure ??.

4 Discussion

In this study, we developed a machine learning-based framework for predicting mortality risk among patients with heart failure. By combining statistical factor identification with machine learning algorithms, clinically meaningful predictors were selected and used to estimate individual mortality probability.

The risk factor analysis identified several important variables associated with mortality. Serum creatinine was positively associated with mortality risk, which may reflect the influence of impaired renal function on heart failure progression. The interaction between cardiac dysfunction and renal impairment, often referred to as cardiorenal syndrome, has been recognized as an important factor affecting prognosis in patients with heart failure.

Ejection fraction showed a negative association with mortality, indicating that patients with reduced cardiac contractile function may experience increased risk of adverse outcomes. Since ejection fraction is a widely used indicator of left ventricular function, its inclusion in prediction models provides clinical interpretability.

Age was also identified as an important predictor. Older patients generally have increased cardiovascular burden and more frequent comorbidities, which may contribute to increased mortality risk.

From the perspective of machine learning performance, XGBoost achieved the best accuracy and F1-score, demonstrating its ability to capture complex patterns within clinical data. Random Forest achieved the highest AUC, indicating strong overall discrimination ability. Logistic Regression showed slightly lower predictive performance but remained valuable due to its transparency and interpretability, which are important considerations in clinical applications.

However, several limitations should be considered. First, this study was based on a relatively small dataset with 299 patients, which may limit model generalizability. Second, the variable time represented follow-up duration and showed strong association with mortality; therefore, its application in real clinical prediction requires careful consideration because it may introduce information related to follow-up status. Future studies using larger prospective cohorts are needed to validate the proposed approach.

5 Conclusion

This study developed and compared three machine learning models for predicting mortality risk in patients with heart failure. Statistical analysis identified important mortality-associated factors, including follow-up time, serum creatinine, ejection fraction, age, and serum sodium. The developed models were able to estimate individual mortality probability based on clinical characteristics. Machine learning approaches may provide valuable support for risk stratification and personalized management of patients with heart failure.

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